Assignment Part 1

Financial Function

Practice Case 1 (The Tech Upgrade):

A startup needs a $120,000 loan for new servers. The bank offers a 7% APR for 4 years. Calculate the quarterly payment.

ANS. =PMT(7/4, 4*4, 120000, 0, 0)

Practice Case 2 (The Balloon):

The startup negotiates to pay only a portion of the loan, leaving a $40,000 "Balloon Payment" due at the very end of the 4 years. What is the new quarterly payment?

ANS. =PMT(7/4, 4*4, 120000, 40000, 0)

Practice Case 3 (The Delivery Van):

A bakery buys a van for $45,000 on a 5-year loan at 6% interest. Calculate the principal paid in the first month.

ANS. =PPMT(6/12, 1, 5*12, 45000, 0, 0)

Practice Case 4 (The Final Payment Variation):

The owner wants to know how much principal is in the very last payment (Month 60). (Hint: It should be almost the entire payment!)

ANS. =PPMT(6/12, 60, 5*12, 45000, 0, 0)

Practice Case 5 (The Student Loan):

A student owes **$25,000** at **3% interest** over **10 years**. How much interest is charged in the **first month**?

ANS. =IPMT(3/12, 1, 10*12, 25000, 0, 0)

Practice Case 6 (The High-Interest Variation):

If the student's interest rate was actually **9%** (private loan), how much more interest would they pay in that **first month** compared to the 3% rate?

ANS. Extra Interest vs 3% loan
=IPMT(9/12, 1, 10*12, 25000, 0, 0) -IPMT(3/12, 1, 10*12, 25000, 0, 0)

Practice Case 7 (The College Fund):

A couple wants to save for their child's education. They invest $2,000 every year for 18 years into a fund earning 6% annually. Calculate the total.

ANS. =FV(6, 18, -2000, 0, 0)

Practice Case 8 (The Inflation Variation): If the couple decides to invest at the beginning of each year instead of the end to combat rising tuition, how much extra will they have? (Hint: Toggle the [type] parameter).

ANS. Extra amount gained

=FV(6, 18, -2000, 0, 1) - FV(6, 18, -2000, 0, 0)

Practice Case 9 (The Business Buyout):

A company offers to buy your patent for $15,000 paid at the end of every year for 5 years. If your "opportunity cost" (rate) is 7%, what is the patent's value today?

ANS. =PV(7, 5, -15000, 0, 0)

Practice Case 10 (The Immediate Payment):

The company offers to pay the $15,000 at the start of each year instead. How does this increase the present value of the deal?

ANS. Extra value gained

=PV(7, 5, -15000, 0, 1) - PV(7, 5, -15000, 0, 0)

Practice Case 11 (The 15-Year Mortgage):

A homeowner has a $250,000 mortgage at 4.5% for 15 years. Calculate the total interest paid over the entire life of the loan (Months 1 to 180).

ANS. =CUMIPMT(4.5/12, 15*12, 250000, 1, 180, 0)

Practice Case 12 (The Second Year):

Using the same loan, calculate the total interest paid only during the second year (Months 13 to 24).

ANS. =CUMIPMT(4.5/12, 15*12, 250000, 13, 24, 0)

Practice Case 13 (The Straight-Line Loan):

A farmer buys equipment for $20,000 at 5% annual interest over 10 years (annual payments). Using the fixed-principal method, find the interest for Year 1.

ANS. =ISPMT(5, 0, 10, 20000)

Practice Case 14 (The Comparison Variation):

Calculate the interest for Year 1 using the standard IPMT function and compare it to the ISPMT result. Which one starts with a higher interest charge?

ANS. Difference between ISPMT and IPMT

=ISPMT(5, 0, 10, 20000) - IPMT(5, 1, 10, 20000, 0, 0)

Assignment Part 2

Zenith Mart Data Retrieval: (The Executive Dashboard)

Goal: Build a master transaction log.

Q1: In the Transaction table, use a lookup to pull the Product_Name based on the Product_ID from the Sales_Transactions sheet.

Q2: Add a column for Supplier_Name. This requires looking up the Supplier_ID in Product_Master and then finding the name in Supplier_Master. How can you do this in one step?

ANS. =XLOOKUP(XLOOKUP(B32, Product_Master!A:A, Product_Master!F:F, "not found", 0),Supplier_Master!A:A,Supplier_Master!B:B,"not found",0)

Q3: Pull the Stock_Status from the Inventory sheet into the Sales report. Which function is most efficient for this?

ANS. =XLOOKUP(A23, Inventory_Status!A3:A14, Inventory_Status!D3:D14, "not found",0)

Q4: A manager wants to see the Net_Revenue alongside the Stock_Status. If a product is "Critical" but has high revenue, it needs an urgent restock. Create this view using the functions learned.

SECTION D — Combined Lookup: Enrich Sales Transactions with Product & Supplier Info						
Transaction_ID	Product_ID	Product_Name	Category	Supplier_Name	Net_Revenue (₹)	Stock_Status
T001	P001					
T004	P009					
T008	P004					
T012	P008					
T020	P012					

ANS.

SECTION D — Combined Lookup: Enrich Sales Transactions with Product & Supplier Info						
Transaction_ID	Product_ID	Product_Name	Category	Supplier_Name	Net_Revenue (₹)	Stock_Status
T001	P001	Basmati Rice 5kg	Staples	AgroFresh Traders	14,725.00	Adequate
T004	P009	Biscuit Pack 250g	Snacks	SnackWave Co.	6,000.00	Adequate
T008	P004	Coconut Oil 500ml	Edible Oils	Kerala Naturals	7,560.00	Low
T012	P008	Ball Pen Box (10pcs)	Stationery	OfficeLine Supplies	3,847.50	Critical
T020	P012	Instant Coffee 100g	Beverages	BevCraft India	4,203.75	Critical